Causal Machine Learning Report

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This project investigates the causal effect of participation in 401(k) retirement savings plan on individual's accumulated assets. A key challenge in estimating this effect is the non-random nature of 401(k) enrollment: individuals with a stronger preference for saving are more likely to participate in such plans and also tend to have higher accumulated assets regardless of participation. As a result, naïve comparisons between participants and non-participants are likely to suffer from confounding due to unobserved savings preferences, leading to biased estimates of the true effect.

To address this challenge, we begin by examining the effect of *eligibility* for a 401(k) plan, rather than actual participation. Figure 1 shows a causal diagram of the situation at hand. Here U denotes unmeasured confounders like saving preferences. Importantly, there is no causal connection between U and 401k eligibility. This is the case since 401k eligibility is assumed to be not influenced by saving preference at this time.

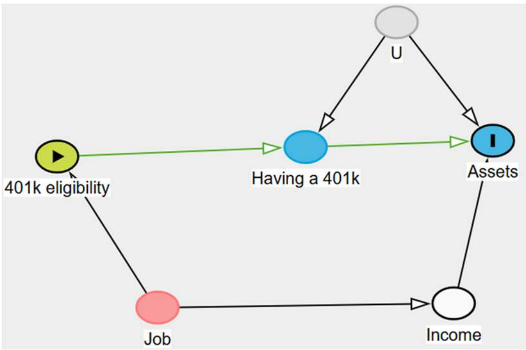

Figure 1: Causal diagram of the situation at hand.

When controlling for income, the backdoor path "401k eligibility ← Job → Income → Assets" is closed. Thus eligibility for a 401k can be taken as exchangeable conditional under income. Specifically, for people with similar incomes, the counterfactual Assets are independent of whether the person had or did not have 401k eligibility. There is only one causal pathway between 401k eligibility and Assets and it goes trough "Having a 401k". We are thus doing a kind of instrumental variable analysis using 401k eligibility as the instrument.

In a first analysis, we used AIPW and TMLE to estimate the average treatment effect of being eligible for a 401k on net financial assets. In this and all further analyses, this was done using 5-fold cross-fitting. Moreover, the SL libraries SL.glm, SL.glmnet, SL.randomForest and SL.earth were used in this and all further analyses. The AIPW estimate was calculated using the npcausal package in R. The estimated nuisance parameters as well as the tmle package were used to obtain the TMLE estimate. A second analysis was performed without covariate adjustments. Here, cross fitting was done manually. The TMLE estimate was obtained by rescaling the Y variables to [0, 1] (which is valid according to (Frank HA, Karim ME, 2023)), then fluctuation models of the form: $\text{logit}\big(E(Y|A)\big) = \text{logit}\left(Q_n^{(0)}\right) + \delta\frac{A}{g_n}$ were used. After which the outcome variables were scaled back to calculate the ATE estimate and its confidence interval.

The results are shown in table 1. When adjusting for the confounding variables, AIPW estimates that on average being eligible for a 401k increases the net financial assets with 8.1 thousand dollars, with a confidence interval of (5.9, 10.3) thousand dollars. TMLE produces a similar estimate, an increase of 8.0 thousand dollars, with a CI of (5.8, 10.3) is expected when being treated. When not adjusting for any confounder, the increase predicted by AIPW and TMLE changes to an increase of 19.6 thousand dollars in net financial assets. If 401k eligibility is positively associated with income and income positively associated with assets, then one expects that indeed not adjusting will result in a bigger ATE. Looking at these results, it is pretty clear that both estimating methods produce very similar results. There is a big difference, however, in not adjusting for confounders.

Table 1: *Average treatment effect of 401 (k) eligibility on net financial assets with and without covariate adjustments using AIPW and TMLE.*

	ATE [1,000 $]	95% CI [1,000 $]
Results with covariate adjustment		
AIPW	8.1	[5.9, 10.3]
TMLE	8.0	[5.8, 10.3]
Results without covariate adjustment		
AIPW	19.6	[16.8, 22.3]
TMLE	19.6	[16.8, 22.3]

In calculating these estimates, it is important that the estimated propensity scores do not become too extreme. The estimated propensity scores were all in the range of [0.05, 0.86]. We assumed that these values were not too extreme and thus there was no need to do different analyses. The ATT was not calculated because the ATE is a metric that is practically more of interest. The ATE carries information on how much assets an individual from the population would gain when being eligible for a 401k. There is no need to deviate from this as the propensities are not extreme.

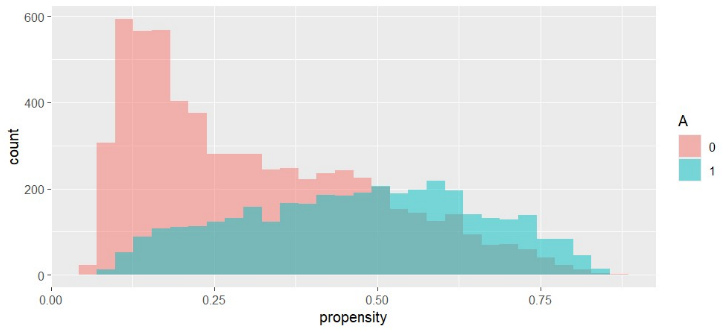

Figure 2: Histogram of the propensity scores by treatment.

Figure 2 shows that the propensity scores also show great overlap between treated and untreated. There is no danger for extreme extrapolation. However, a boxplot of the EICs (figure 3) shows the presence of a lot of outliers and some very heavy outliers. Taking these two findings into account means that the extreme values are likely due to a weak model fit and not due to instability in the propensity scores. This may cause the CIs to be unreliable.

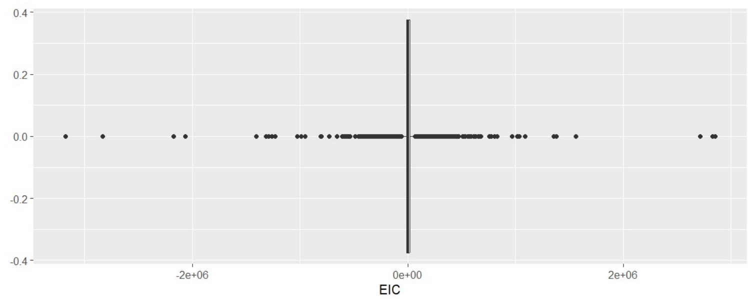

Figure 3: Boxplot of the efficient influence curves for the AIPW estimates with covariate adjustments.

Table 2 shows the results of a second analysis. Here we did a more powerful analysis to test if eligibility for a 401k effects the assets. The null hypothesis states that the expected covariance, $\theta = 0$. This Null hypothesis cannot be rejected on a confidence level of 5%. This test fails to reject the null hypothesis, so we cannot conclude that eligibility affects the net financial assets. This is in contrast to the results in table 1. This is yet another reason (besides figure 3) to take these results with a grain of salt.

Table 2: *Expected covariance and related estimates.*

	Estimate	SE	95% CI	p-value
Theta	76	269	[-450, 602]	0.78

Next, conditional treatment effect were calculated using an R-learner. These were calculated manually to permit a fully cross-fitted implementation, as well as the use of different complementary algorithms. Of note, as the cross-fitted approach was computationally intensive, we used SL.ranger instead of SL.randomForest for this analysis. Notably, inspection of the pseudo-outcomes revealed outliers. Similarly to previous analyses. further investigation indicated that these were primarily due to poor outcome predictions, rather than instability in propensity scores. As can be seen in Figure 4, the conditional average treatment effect seem to increase with income, but not age and years of education, after removing an outlier.

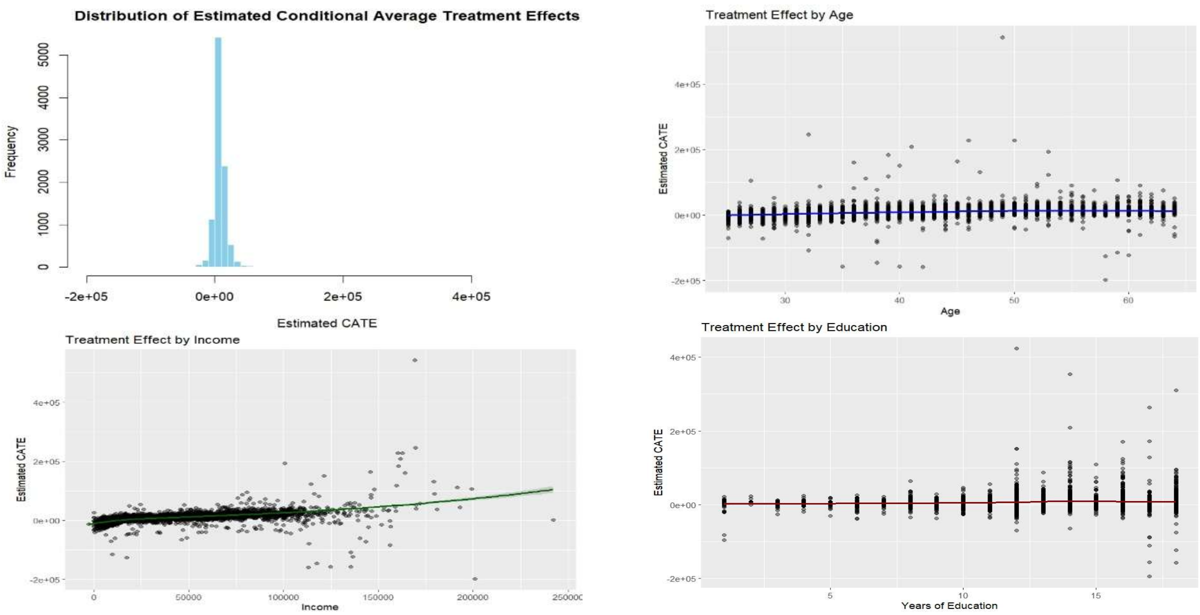

Figure 4: Distribution of conditional average treatment effects, and treatments effects by age, income and education

Hereafter the counterfactual pseudo-outcomes were estimated using the DR-learner. Figure 5 shows these in a histogram. The expected counterfactual prediction error was computed using again AIPW estimation, the result is shown in table 3. So one expects that on average the squared difference between the pseudo-Y^1 and the counterfactual Y^1 equals 3059 thousand dollars squared (a million squared dollars).

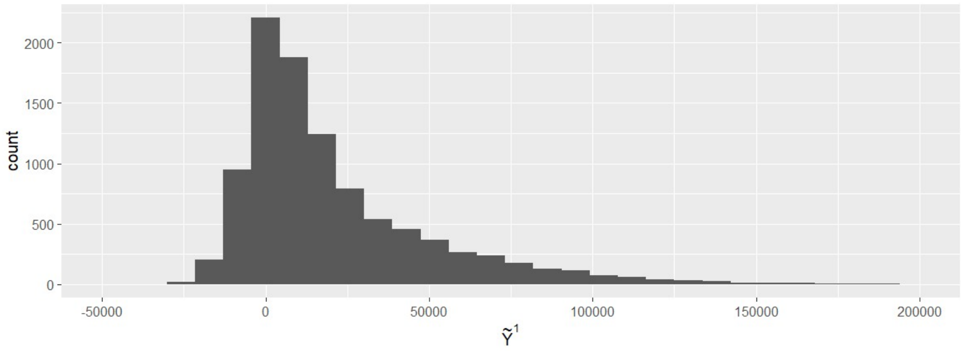

Figure 5: Histogram of the pseudo-outcomes.

Table 3: *Results of the AIPW estimation of the mean squared prediction error.* C^1 denotes the counterfactual pseudo-outcome.

	[(1,000$)²]	95% CI [(1,000$)²]
$E[(Y^1 - C^1)^2]$	3059	[1821, 4298]

However, these results are to be taken with a grain of salt. Figure 6 shows the predicted counterfactual prediction error. Here, the plot shows some very clear outliers that probably skew the result and bias the confidence intervals. Similarly, the EICs have some pretty high outliers (a plot is not shown, but it looks very similar to figure 6). Again a weak model fit is probably the root cause of this problem.

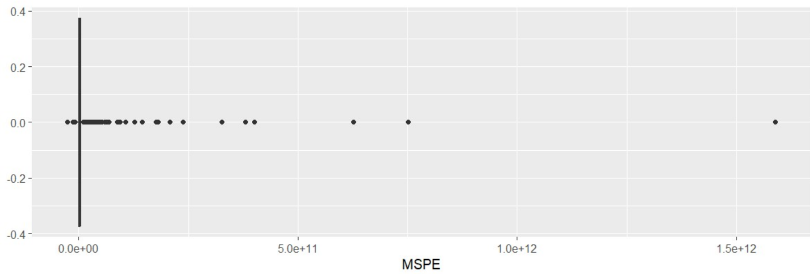

Figure 6: Boxplot of the estimated MSPEs .

Task distribution

Exercises 1-3 and 5 were mainly done by Lennert, exercise 4 was done by Marie.

References

Frank HA, Karim ME. Implementing TMLE in the presence of a continuous outcome. Research Methods in Medicine & Health Sciences. 2023;5(1):8-19. doi:10.1177/26320843231176662